CHEMICAL BONDING

↳ Force of attraction present between any chemical species like, atoms, ions & Molecules.

Bond formation → Exothermic

Bond dissociation → Endothermic

Chemical bonds on the basis of Bond Energy

Strong bond	Weak bond
① Ionic bond	① Hydrogen bond
② Covalent bond	② Vander waal's bond
③ Co-ordinate bond	
④ Metallic bond	

Cause of chemical combination

Modern concept :- Every atom has a tendency to form Max^{x}. possible bonds in order to minimize its potential energy & gain stability.

Classical concept (octet rule) :-
↳ Every atom has a tendency to acheive stable outer octet (8e$^-$ in valence shell).

Different theories to explain bonding

① Octet rule

② VSEPR (valence shell electron pair repulsion theory)

③ VBT (valence bond theory)

④ MOT (Molecular orbital theory)

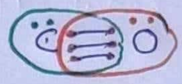
Kossel-lewis Approach (octet rule):

- Based on inertness of noble gas
- Atoms can be combine either by transfer or by sharing of valence electrons in order to have an octet (8e- in their valence shells).
- In case of H & He duplet (2e$^-$ in valence shell) will be completed.

How to write Lewis dot structure & bond formation of Molecules.

$$\text{No. of Bonds} = \frac{\overset{(\text{No of atoms} \times 8)}{\text{Req. valence } e^-} - \text{Available val. } e^-}{2}$$

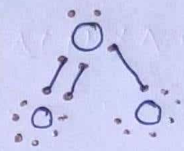
E.g → $\boxed{CO}$



No. of bonds b/w C & O $= \frac{(8\times2)-(4+6)}{2} = 3$

$\boxed{O_3}$

No. of bonds $= \frac{(8\times3)-(6\times3)}{2} = 3$



$\boxed{N_3^-}$

No. of bonds $= \frac{8\times3-(5\times3)+①}{2} = 4$

$:\ddot{N}=N=\ddot{N}:$

$N\equiv N-\ddot{N}:$

Limitations of octet rule

(i) Existence of Hypervalent compound.

$\boxed{\text{val } e^- > 8}$ E.g → PCl_5, SF_6, H_2SO_4.

(ii) Existence of Hypovalent compound.

$\boxed{\text{val } e^- < 8}$ E.g → LiCl, BF_3, $BeCl_2$.

(iii) odd electron species donot follow octet rule. Eg → NO, NO_2.

(iv) Existence of compounds of inert gases with oxygen & fluorine like XeF_2, KrF_2, $XeOF_2$ etc.

(v) does not explain the relative stability of the molecules being totally silent about the energy of a molecule.

(vi) This theory does not account for the shape of molecules.

⇒ Formal charge :-

F.C on atom = Jitne hone chahiye (val e$^-$ in free state) − Jitne hain (val e$^-$ in given structure)

$\boxed{F.C = \text{Total val. } e^- - [2 \times l.p + \text{Bond pair}]}$

Eg → ① O_3

(6-5) = +1

(6-6) = 0 (6-7) = -1

⇒ Stability of same molecules of diff str

① More stable due to symmetrical charge distribution.

② More stable due to absence of charge.

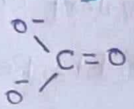
③ More stable due to -ve charge present on high EN element.

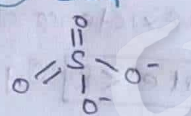
Avg. formal charge & Avg. X-O bond order in oxo-Anions:-

Avg. FC = $\frac{\text{total charge present on O-atom}}{\text{No. of O-atom where -ve charge can be placed}}$

Avg. X-O bond order = $\frac{\text{total no. of X-O bonds}}{\text{no. of X-O pairs participate in resonance}}$

① CO_3^{2-} ② SO_4^{2-}





	① CO_3^{2-}	② SO_4^{2-}
F.C on 'O' →	-2/3	-2/4
X-O bond → order	4/3	6/4

Trick	FC
=O →	0
-O →	-1
≡O or =O- →	+1

Covalent bond

↳ Formed by the equal sharing of e^-.

★ Covalency:- Total no. of covalent bond (σ & π) formed by an atom in a molecule.

VALENCY (combining capacity)

① Covalency → (Total no. of σ & π bonds)

② Electrovalency → Combining capacity in ionic comp. Eg → Na^+Cl^- → $Na^+ = +1$, $Cl^- = -1$

NOTE:-

① Max covalency of 2nd period element can not be more than '4' due to absence of d-orbitals.

② 3rd period elements can extend their covalency beyond '4' by using vacant 3d orbitals.

③ 'F' can show only 1 covalency.

Coordinate bond

↳ Formed by the unequal sharing of e^-.

↳ formed b/w e^- donor & e^- acceptor.

$A \rightarrow B$ σ coordinate

or

$\overset{+}{A} - \overset{-}{B}$

↳ e^- donor = Lewis base

↳ e^- acceptor = Lewis Acid

Coordinate

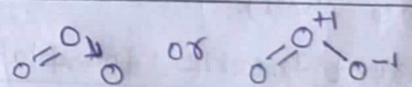
σ: $A \rightarrow B$

π: X—Y π coordinate (Back bonding) → Non included in covalency.

π Coordinate $\overleftarrow{C \equiv O}$ or $\overset{-}{C} \equiv \overset{+}{O}$

Species in which σ coordinate bond (dative bond) is present.

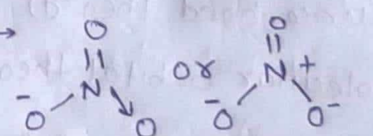
① O_3 →



② NH_4^+ → $[\ddot{N}H_3 \rightarrow H^+]$

③ H_3O^+ → $[H_2\ddot{O} \rightarrow H^+]$

④ NO_3^- →



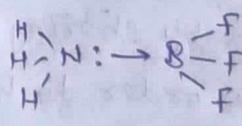
-2/3

4/3

⑤ PCl_6^- → $(Cl^- \rightarrow PCl_5)$

⑥ BF_4^- → $F^- \rightarrow BF_3$

⑦ $NH_3 \cdot BF_3$

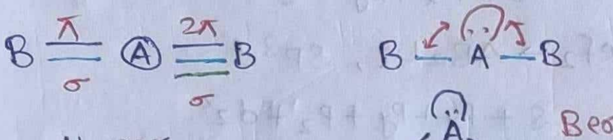


⑧ PH_4^+ $[H_3P: \rightarrow H^+]$

⑨ $B(OH)_4^-$ → $B(OH)_3 \leftarrow \ddot{O}H^-$

VSEPR (Valence shell electron pair Repulsion theory):—

- Based on the repulsive interactions of the electron pairs in the valence shell of the atoms.
- Overall geometry/shape of molecule depend on 'σ' bond pairs & lone pairs.



linear Bent

Order of Repulsion

$\boxed{l.P - l.P > l.P - B.P > B.P - B.P}$

Types of Monocentric Molecule

σ-bond Lone pair

central Atom ← $\boxed{A\ B_n\ L_m}$ → no. of L.P

no. of B.P

Monocentric Molecules		Shape/Geometry
$\boxed{AB_2L_0}$	⟶	Linear
$\boxed{AB_3L_0}$	⟶	Trigonal planar
AB_2L_1	⟶	Bent / V-shape
$\boxed{AB_4L_0}$	⟶	Tetrahedral
AB_3L_1	⟶	Trigonal Pyramidal
AB_2L_2	⟶	Bent / V-shape / Angular
$\boxed{AB_5L_0}$	⟶	Trigonal Bipyramidal
AB_4L_1	⟶	See-Saw
AB_3L_2	⟶	Bent 'T'-shape
AB_2L_3	⟶	Linear
$\boxed{AB_6L_0}$	⟶	Square Bipyramidal or Octahedral
AB_5L_1	⟶	Square pyramidal
AB_4L_2	⟶	Square planar
$\boxed{AB_7L_0}$	⟶	Pentagonal Bipyramidal
AB_6L_1	⟶	Distorted octahedral
AB_5L_2	⟶	Pentagonal planar

⇒ How to find Hybridisation

(L.P+σ) (M+n)		Type of Hybridisation
2	⟶	SP (S+P)
3	⟶	SP^2 (S+P+P)
4	⟶	SP^3 (S+P+P+P)
5	⟶	SP^3d (S+P+P+P+d) etc.

⇒ Trick to find n & m in AB_nL_m:—

$\boxed{\dfrac{\text{Total valence } e^-}{8}}$ Quotient = n → no. of σ-bond

$\dfrac{\text{Remainder}}{2}$ = m → no. of l.P

① BF_3 $\dfrac{3+7+7+7}{8} = \dfrac{24}{8}$ Q = 3(n) R = 0

AB_3L_0, SP^2, Trigonal planar

② NO_2^+ $\dfrac{5+6+6-1}{8} = \dfrac{16}{8}$, Q = 2 R = 0

AB_2L_0, linear, SP

③ NO_3^- $\dfrac{5+(6\times3)+1}{8} = \dfrac{24}{8} = Q = 3$ R = 0

AB_3L_0, SP^2, T. planar

④ SO_2 $\dfrac{6+6+6}{8} = \dfrac{18}{8}$ Q = 2(n) R = 2 R/2 = 1(m)

AB_2L_1, Bent SP^2

NOTE:- Do not apply this trick for odd electron species & species having hydrogen as S.A.

⇒ If H & Halogens are S.A (Monovalent)

↳ Jitne hai utne C.A ke valence e^- kam kr do!!, Bache huye ke l.P bna do.

① CH_4

valence e^- of C = 4
no. of H = 4
l.P = 4−4 = 0

AB_4L_0, SP^3 tetrahedral

② NH_4^+

valence e^- of N = 5
= (5−1) = 4
(∵ + hai)

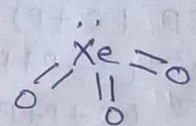
AB_4L_0

③ NH_2^-

→ (5+1) = 6
6−2 = 4 = 2.l.P
AB_2L_2

NOTE:- Jab compound me koi bhi charge nahi Ho, aur S.A oxygen ho to sare "O" double bonded consider honge.

E.g → XeO_3



Hybridisation

- Mixing of orbitals
- Jitne milenge, utne Banenge
  (A.O) (H.O)
- H.O are equivalent in shape & Energy.
- Hybrid orbital ke 2 kaam
  ↓ [no. of H.O = σ + l.p]
  σ Banayega & l.p Rakhega
- Size of H.O → $SP < SP^2 < SP^3$
  % P ↑ size ↑
- 'S' Angle ki dukaan hai.
  'P' direction ki.

⇒ **Mixing hogi to**
- S ka directional ↑se
- P ka angular Nature ↑se

Bond Angle :- Gen. 1 LP ↓se B.A by 2.5° to 2.8°.

- L.P lagao, Angle ghatao.

→ SP > SP^2 > SP^3

SP	SP^2	SP^3
% S = 50%	% S = 33%	% S = 25%
θ = 180°	θ = 120°	θ = 109.5°

If Hybridisation same

If L.P is at C.A
- L.P ↑, B.A ↓
  ↓
  If hybrid. & L.P both are same
  → Size of S.A ↑, B.A ↑
  → Size of C.A ↑, B.A ↓

If L.P is absent at C.A
- give ur answer A1q to hybridisation, if hybr. same, Angle same
  $BF_3 = BCl_3$
  AB_3L_0 AB_3L_0
  SP^2 SP^2

Types of d-orbitals used in hybridis.

$SP^3d \rightarrow dz^2$
$SP^3d^2 \rightarrow dz^2, dx^2-y^2$
$SP^3d^3 \rightarrow dz^2, dx^2-y^2, dxy$

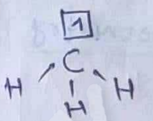
Eg → SF_6 → AB_6L_0, SP^3d^2
$S + P_x + P_y + P_z + dz^2, dx^2-y^2$

Eg → XeF_2 → AB_2L_3, SP^3d
$S + P_x + P_y + P_z + dz^2$.

NOTE:- orbitals remains pure (not used in hybridisation).
E.g → (dyz, dxz)

Hybridisation in Odd e⁻ Species

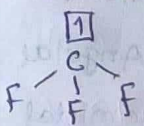
- Jab S.A 'H' Hoga, hybridisation me odd e⁻ vala dabba participate nahi karega. Eg → CH_3



∵ EN of S.A < E.N of C.A.
→ SP^2

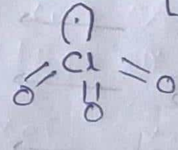
- orbital having odd e⁻ can participate in hybridisation only when S.A is high EN.

E.g → CF_3



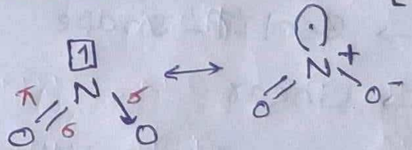
SP^3

ClO_3 [$EN_O > EN_{Cl}$]



SP^3

⇒ NO_2 (V. Imp) [$E.N_{S.A} > EN_{CA}$]



- Hybridisation of N = SP^2
- Shape – Bent
- O–N–O bond Angle 120° < θ < 180°
  θ ≃ 134°
- ~~Unpaired~~ 1σ coordinate bond is present.
- Acidic oxide
- can cause Acid rain
- Brown coloured gas, Paramagnetic
- can demerise into N_2O_4 in High P & Low T.

Hybridisation of some molecules in solid state :- Exist in form of ions.

↳ PCl_4^+ | PCl_6^- * PBr_6^-

AB_4L_0 AB_6L_0 ↳ donot exist due to
tetra   octahedral   large size of Br atom
sp^3 sp^3d^2 (steric hindrance)

PBr_4^+ ✓

Existing & Non-existing Molecules

(A) Non - contraction of d-orbitals

Existing	Non-existing
XeF_6 (sp^3d^3)	XeH_6
XeF_2 (sp^3d)	XeH_2
SF_6 (sp^3d^2)	SH_6

↳ When S.A is high EN, Zeff of C.A ↑se.
↳ If C.A have to use its d-orbitals in hybridisation they must be sufficiently contracted it is only possible when S.A is high EN.

(B) Improper covalency

Existing		Non-existing
NCl_3	2nd period elm	NCl_5
BF_4^-	Max. covalency = 4	BF_6^{3-}
ClF_3	F = only one	FCl_3

(C) Steric Hindrance

Existing	Non-existing
PCl_6^-	PBr_6^-
SiF_6^{2-}	$SiCl_6^{2-}$
IF_7	ICl_7 / IBr_7

Types of overlapping

↳ Always σ > π (strength)
∵ Co-axial overlapping > collateral

• Co-axial / Head-on overlapping
↳ Along with Internuclear axis (INA)
↳ σ Bond formation

• Collateral / Sideways overlapping
↳ perpendicular to INA.
↳ π Bond formation

① H.O + H.O (C.A) (S.A) → Always σ.
H.O + P.O (C.A) (S.A) → Always σ.

② P.O + P.O → σ/π

Types of P.O + P.O overlapping (σ/π)

① s-s ② s-p ③ p-p
④ p-d ⑤ d-d

① s-s   (INA) → s kvi π-bond nhi bna skta h.

s, s: x → σ, y → σ, z → σ

② s-p

s, p(x): x → σ, y → 0, z → 0

→ common par σ, baaki par zero.

③ p-p

$p_x + p_x$: x → σ, y → π, z → π

→ common par σ, baaki par π.

NOTE → Both 'p' must be same.

$p_x + p_y$: x → 0, y → 0, z → 0

④ p-d

$d_{xy} + p_x$: x → 0, y → π, z → 0

→ Uncommon par π, baki par zero.

$p_x + d_{yz}$: x → 0, y → 0, z → 0

⑤ d-d

d_{xy}, d_{xy}: x → π, y → π, z → δ

→ common par π, baaki par δ

NOTE → Both 'd' must be same.

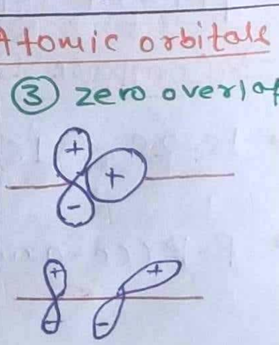
d_{xz}, d_{yz}: x → 0, y → 0, z → 0

Overlapping of pure Atomic orbitals

① Positive overlap
⊕ + ⊕ , ⊖ - ⊖ ,

② Negative overlap
⊕ + ⊖

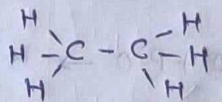
③ zero overlap



Bond Identity

① σ-Bond Indentity

ⓐ H−H (S−S) ⓑ H−Cl (S−P)

ⓒ Cl−Cl (P−P) ⓓ 

C−C (sp^3-sp^3)

C−H (sp^3-1s)

② π-bond Identity (imp)

NOTE → If C.A belongs to 2^{nd} period then there is no 'd' available, so no dπ-dπ bonds.

ⓐ SP → All are Pπ-Pπ

ⓑ SP^2 → first one is Pπ-Pπ, Rest are dπ-dπ

ⓒ SP^3 or any other → All are dπ-dπ.

Bond strength / Bond Energy

(i) Bond strength $\propto \frac{1}{\text{Size (Shell no.)}}$

1S-1S > 2S-2S > 3S-3S

(ii) If size same:-

(a) S−S < S−P < P−P (for co-axial overlapping)

2S-2S < 2S-2P < 2P-2P

$\boxed{\text{B.S} \propto \text{directional nature}}$

(b) Pπ-Pπ < Pπ-dπ < dπ-dπ (for collateral overlapping)

3Pπ-3Pπ < 3Pπ-3dπ < 3dπ-3dπ

$\boxed{\text{B.S} \propto \text{extend of overlapping}}$

(iii) Lone pair-lone pare repulsion can show exceptions only in 2^{nd} period elements. (only in single-bonded atom)

$\boxed{\text{l.P-l.P repulsion} \downarrow \text{se B.S}}$

⇒ Compare Bond strength in following

- 1S-2S < 1S-2P (directional nature of P ↑ B.S ↑)
- P-P (co-axial) > P-P (collateral)
  σ π

- N-N < P-P
- O-O < S-S]→ (l.P - l.P repulsion)

- $Cl_2 > Br_2 > F_2 > I_2$
- 1S-1S > 2P-2P > 2S-2P > 2S-2S > 3S-3S
- 3Pπ-3Pπ < 3Pπ-3dπ < 3dπ-3dπ
- 2Pπ-2Pπ > 3dπ-3dπ > 3Pπ-3dπ > 3Pπ-3Pπ

Dipole Moment (μ) :- Measure overall polarity in Molecule

Unit = Debye

1 Debye = 3.3×10^{-30} cb.m

$\boxed{\mu = q\times d}$ μ = 0 (Non-polar)
μ ≠ 0 (Polar)

- vector quantity
- directed towards more EN elements. (More e^- density)
- direction of l.P dipole moment is always away from the molecule.

For All symmetrical structures

AB_2L_0 to AB_7L_0	Non-polar
AB_4L_2, AB_2L_3	Always μ = 0
	S.A must be same.

⇒ Compare dipole Moment

① $CO_2 < SO_2$
AB_2L_0 AB_2L_1
μ=0 μ≠0

② $SO_2 > SO_3$
AB_2L_1 AB_3L_0
μ≠0 μ=0

③ $CCl_4 < CH_2F_2$
AB_4L_0 AB_4L_0
μ=0 μ≠0
∵ SA diff.

④ $XeF_2 = XeF_4$
AB_2L_3 AB_4L_2
μ=0 μ=0

⑤ $NH_3 > NF_3$
AB_3L_1 AB_3L_1
μ≠0 μ≠0

⑥ $H_2O > OF_2$
AB_2L_2 AB_2L_2
μ≠0 μ≠0
same

↳ In NH_3, l.P dipole moment & Bond pair dipole moment are in same direction. ∴ μ↑

⇒ Examples based on 'q' (ΔEN ↑ 'q' ↑ μ ↑)

① HF > HCl > HBr > HI

② HF > NH_3 > PH_3

③ $H_2O > NH_3 > PH_3$ ④ $H_2O > H_2S$

⇒ Some data based examples

① $CH_3Cl > CH_3F > CH_3Br > CH_3I$

② $CH_3Cl > CH_2Cl_2 > CHCl_3 > CCl_4$

③ $H_2O > HF$ (Acc. to NCERT)

⇒ Calculation of Ionic character

$$\boxed{\% \, IC = \frac{\mu_{Real}}{\mu_{Imaginary}} \times 100} \qquad \mu = q \times d$$

Q→ If $\mu_{HCl} = 1.02D \rightarrow 1.02 \times 3.3 \times 10^{-30}$ C.m

$d_{HCl} = 1.27 A^\circ \rightarrow 1.27 \times 10^{-10}$ m.

Find % IC in HCl?

Ans:- $\% \, IC = \dfrac{1.02 \times 3.3 \times 10^{-30}}{\underset{(q)}{1.6 \times 10^{-19}} \times \underset{(d)}{1.27 \times 10^{-10}}} \times 100$

Ionic Bond / Electrovalent Bond

- formed by complete transfer of electron
- Electrostatic / coulombic force of attraction b/w oppositely charged ions (q^+ & q^-).

$$\boxed{F = \frac{K q_1 q_2}{r^2}}$$

- Generally formed b/w Metals & Non-metals
- Generally solid, brittle in nature at room temp.
- Do not show isomerism (due to presence of directional Bond)
- Polar & Readily soluble in polar solvent like water.

$\boxed{NH_4Cl}$ → covalent (NH_3)
coordinate ($\ddot{N}H_3 + H^+$)
Ionic (NH_4^+ { Cl^-)

Hydration :- Break into ions.

↳ Tendency of Ionic compounds.

$$\boxed{\text{Degree of Hydration} \propto \text{Ionic potential} \text{ or } \text{Hydration Energy}}$$

$$\boxed{\text{Ionic potential}(\phi) = \frac{\text{charge}}{\text{Radius}}}$$

$Li^+ < Be^{2+}$ (ϕ)

Li^+	Na^+	K^+	Rb^+	Cs^+
Ⓐ	Ⓑ	Ⓒ	Ⓓ	Ⓔ

→

$\phi \downarrow$, Ionic size ↑, (degree of hydration) ↓

(hydrated ion size) ↓, Ionic Mobility or Conductance ↑

⇒ Ionic compounds have M.P & B.P as compare to covalent compounds.

$\boxed{\text{Ionic compounds}}$

- Exist in form of Ionic lattice.
- Donot exist in Molecular form due to strong electrostatic force of attraction.

★ Electrical conductivity of I.C

	Ion	Mobility	Conductance
Solid	✓	✗	✗
Liq (Molten)	✓	✓	✓
~~Aqueous~~ Aquous (In form of hydrated ion)	✓	✓	✓

NOTE → Metals are always conductor of electricity due to presence of free e^-.

$\boxed{\text{Covalent compounds (Molecular Solid)}}$

- Exist in form of Molecules & weak vanderwaal force present b/w molecules.

★ Electrical conductivity

	Ions	Mobility	Conductance
Solid	✗	✗	✗
Liq (Molten)	✗	✗	✗
Aquous	✓	✓	✓

★ To form stable compound :-

Released Energy (Profit) > Absorbed energy (Investment)

EXO > ENDO

E.A, L.E > S.E, I.E, B.D.E

Lattice Energy (U) :- Energy req. to break 1 Mole Ionic solid into its gaseous constituents ions.

$$\boxed{U \propto \frac{|q_1| |q_2|}{r}}$$ $|q| \uparrow, U \uparrow$

If charge is same $\boxed{U \propto \frac{1}{r}}$ $r \uparrow U \downarrow$

Eg →
$\overset{+1}{Na}\overset{-1}{Cl} < \overset{+1}{Na}\overset{-1}{F}$
$\overset{+2}{Mg}\overset{-2}{O} < \overset{+3}{Al_2}\overset{-2}{O_3}$
$\overset{+1}{Li_3}\overset{-3}{N} > \overset{+1}{Na_3}\overset{-3}{N}$

NOTE:-

AM → Only Li can form stable nitride

AEM → All can form stable nitride. due to high L.E (high charge & small size)

Eg → Be_3N_2, Mg_3N_2

Fajan's Rule → Development of covalent character in ionic compounds.

→ Cation have more Zeff than Anion.
→ Distortion in e^- cloud of Anion
→ Merging e^- clouds of cation & Anion.

$\boxed{\text{Polarisation} \uparrow, \text{covalent character} \uparrow}$

- Polarising Power ∝ Polarisation (ϕ) (for cations)
- Polarisability ∝ Polarisation (for Anions)

Condition for Max. covalent character

→ $r^+ \downarrow$ (small cation) $Be^{2+} > Mg^{2+} > Ca^{2+} > Sr^{2+}$
→ $r^- \uparrow$ (Large Anion) $F^- < Cl^- < Br^- < I^-$
→ $q^+, q^- \uparrow$ (high charge on ions)

Eg → $M^+ < M^{2+} < M^{3+}$

$X^- < X^{2-} < X^{3-}$

Important points

$$\boxed{\phi \text{ for cation} = \frac{q^+ \text{(charge)}}{r^+ \text{(radius)}}} \quad r^+ \uparrow \ \phi \downarrow$$

(i) Na^+, K^+, Rb^+, Cs^+ = Low ϕ

(ii) D-block cations ϕ > S-block cations (size ↓, Zeff ↑, ϕ ↑)

(iii) $LiCl, BeCl_2, MgCl_2$ soluble in organic solvents like benzene, pyridine and Acetone. (Non-polar solvent)

covalent ch ↑ due to very high ϕ nature

(iv) Due to covalent character, M.P of ionic compounds decreases.

$\overset{+2}{Sn}Cl_2 < \overset{+4}{Sn}Cl_4$ (cov. ch.)

$SnCl_2 > SnCl_4$ (M.P)

Q → Compare covalent character

① $LiF < LiCl < LiBr < LiI$ ($r^- \uparrow$)

② $LiCl > NaCl > KCl > RbCl$ ($r^+ \downarrow$)

③ $\overset{+2}{Sn}Cl_2 < \overset{+4}{Sn}Cl_4$ (high charge on ions)

④ $NaCl < CuCl$ (d-block > s-block)

Molecular orbital theory

- Paramagnetic behavior of O_2 is explained by MOT.
- Applied only for homodiatomic molecules (total electrons upto 20).
- The Molecular orbitals like A.O are filled in accordance with the aufbau principle obeying the Pauli's exclusion principle & the Hund's rule.
- A/q to MOT

A.O + A.O → ABMO (high energy) Antibonding MO.
A.O + A.O → BMO (low energy) Bonding M.O.

Electronic configuration (INA = Z)

- Beyond N_2 (1221) [Without S-P mixing] (Normal)

$$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < \begin{matrix} \pi 2p_x \\ \pi 2p_y \end{matrix} < \begin{matrix} \pi^* 2p_x \\ \pi^* 2p_y \end{matrix} < \sigma^* 2p_z$$

1 2 2 1

- Upto N_2 (2121) [With S-P Mixing] (Abnormal)

$$\text{,,} \quad \text{,,} < \begin{matrix} \pi 2p_x \\ \pi 2p_y \end{matrix} < \sigma 2p_z < \begin{matrix} \pi^* 2p_x \\ \pi^* 2p_y \end{matrix} < \sigma^* 2p_z$$

2 1 2 1

Eg →

① O_2 (1221) → 16 e^-

$\sigma 1s^{\uparrow\downarrow}$ $\sigma^* 1s^{\uparrow\downarrow}$ $\sigma 2s^{\uparrow\downarrow}$ $\sigma^* 2s^{\uparrow\downarrow}$ $\sigma 2p_z^{\uparrow\downarrow}$ $\pi 2p_x^{\uparrow\downarrow}$ $\pi 2p_y^{\uparrow\downarrow}$ $\pi^* 2p_x^{\uparrow}$ $\pi^* 2p_y^{\uparrow}$ $\sigma^* 2p_z$

$$B.O = \frac{10-6}{2} = 2$$

2 unpaired e^-
∴ O_2 is paramagnetic

② N_2 (2,7) → 14e^-

$\sigma 1s^2\ \sigma^* 1s^2\ \sigma 2s^2\ \sigma^* 2s^2\ \pi 2p_x^2 = \pi 2p_y^2\ \sigma 2p_z^2\ \pi^* 2p_x = \pi^* 2p_y\ \sigma^* 2p_z$

$B.O = \frac{6-0}{2} = 3$

$$\boxed{\text{Bond order} = \frac{BMO\,e^- - ABMO\,e^-}{2}}$$

- B.O ↑, stability ↑, B.S ↑, B.L ↓

NOTE → Species having same bond order then ABMO e^- ↑, stability ↓se

Super NOTE :-

Total e^- 1 to 8

odd number → 0.5 B.O

even number → 2, 6 → 1 B.O ; 4, 8 → 0 B.O

Total e^- 14 → 3 B.O

8	9	10	11	12	13	14	15	16	17	18	19	20
0	0.5	1	1.5	2	2.5	3	2.5	2	1.5	1	0.5	0

← 1 e^- ↓se in BMO, B.O ↓se by 0.5 | 1 e^- ↑se in ABMO, B.O ↓se by 0.5 →

Magnetic Behavior

Count total e^- → odd (Paramagnetic) ; even (Diamagnetic)

Except → B_2, O_2, S_2 (Paramagnetic)

LCAO (Linear combination of A.O)

(formation of M.O) INA = 2

S–S (σ) → $\sigma 1s, \sigma^* 1s, \sigma 2s, \sigma^* 2s$

P–P (σ) → $\sigma 2p_z, \sigma^* 2p_z$

P–P (π) → $\pi 2p_x, \pi 2p_y, \pi^* 2p_x, \pi^* 2p_y$

NOTE

- All 'σ' MO (BMO & ABMO) are symmetrical Along the bond axis.
- All 'π' MO (BMO, ABMO) are unsymmetrical along the bond axis.

⇒ Magnetic Moment → Jitna unpaired e^- utna point something

(n)	
1	1. ...
2	2. ...
3	3. ...
4	4. ...

$\mu = \sqrt{n(n+2)}$

$= \sqrt{5(5+2)} = \sqrt{35}$.

Hydrogen Bonding :- when hydrogen is linked with high EN element like F/O/N molecule acquire polarity & two adjacent molecules attract each other by H-bonding

$\overset{+\delta}{H} - \overset{-\delta}{X} \cdots\cdots \overset{+\delta}{H} - \overset{-\delta}{X}$ (H-bonding) $\boxed{X = F/O/N}$

- l.p is essential for H-bonding
- H-Bonding is a special case of dipole-dipole interaction.

★ Classification of H-Bonding

InterMolecular H-Bonding	Intramolecular H-bonding
– present b/w molecules	– Present within molecules
– chelation (ring) may occurs.	– chelation always occur
– More association b/w Molecules	– Less association
– High M.P & B.P	– Low M.P & B.P
– Less volatile (low v.p)	– More volatile
– More viscocity	– Less viscosity.

★ Important points about H-bonding

- cellulose can show H-bonding with water.
- Amino acid can show H-bonding with H_2O.
- water is liquid at room temp. while H_2S is gas.
- Fire due to petrol can not be extinguish by water.
- Fire due to Alcohol can be extinguish by water.
- Isomeric ethers are more volatile than Alcohol.
- High B.P & viscocity can be explained by H-bonding.

↳ orth - Nitrophenol is more volatile than para-nitrophenol.

↳ ortho-salicylaldehyde is liquid at room temp. while para-salicyladehyde is solid.

* Order of B.P & M.P in group 15,16,17 hydrides:-

↳ M.P & B.P of group 15, 16, 17 hydrides is mainly depends on weak vandorwaal force (w.vf ∝ M.w) but M.P & B.P of (NH_3, H_2O & Hf) is higher than expectation in their respective group due to presence of H-bonding.

(1>2>3>4)

B.P (2,1,1)

2 NH_3	1 H_2O	1 Hf
4 PH_3	4 H_2S	4 HCl
3 AsH_3	3 H_2Se	3 HBr
1 SbH_3	2 H_2Te	2 HI

M.P (1,1,2)

1 NH_3	1 H_2O	2 Hf
4 PH_3	4 H_2S	4 HCl
3 AsH_3	3 H_2Se	3 HBr
2 SbH_3	2 H_2Te	1 HI

$BiH_3 > SbH_3 > NH_3 > AsH_3 > PH_3$

B.P → $H_2O > Hf > NH_3$

M.P → $H_2O > NH_3 > Hf$

H-bonding strength = $Hf > H_2O > NH_3$

($\Delta EN\uparrow$, $|q|\uparrow$, H-bond↑)

H-bonding in ice

↳ 3d H-bonding $\uparrow d = \frac{M}{V}\downarrow$

↳ open cage like structure (V↑ D↓)

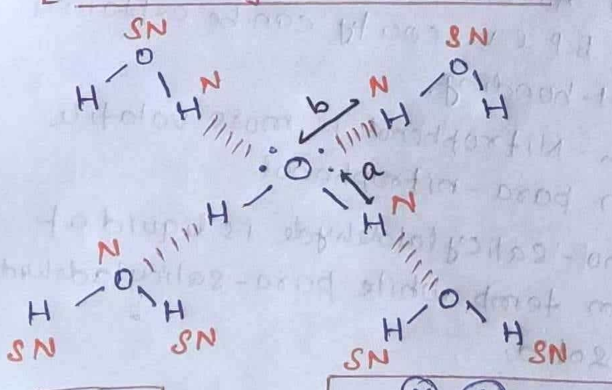
↳ on heating ice contracts (Melts).

∴ ice floats on water ← V↓ density↑

∴ $d_{water} > d_{ice}$

↳ 'O' atom is surrounded by '4' equidistance 'O' atoms.

[H-bonding from 4 sites]



b > a

H-bond B.L   covalent B.L

	N	SN	Total
O	2	2	4
H	2	4	6

$H_2O(s)$ → Open cage like structure

$H_3BO_3(s)$ → 2-D sheet like

$Hf_{(s)}$ → zig-zag

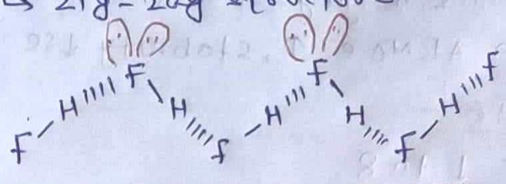
$KHCO_3(s)$ → Anionic dimer

$NaHCO_3(s)$ → Anionic polymer

↳ Max^m no. of H-bonds by H_2O in ice is 4.

H-bonding in Hf Molecules

↳ zig-zag structure.



Dipole Based forces

↳ Interparticle forces ∝ M.W

Ions involved

(1) Ion-dipole (ID)

↳ Present b/w Ions & Polar (Na^+, H_2O)

(2) Ion-Induced dipole (IID)

↳ Present b/w ions & Non-polar (Na^+, Br_2)

Ions donot involved (Vanderwaal forces)

(A) Dipole-Dipole (keesom force) (DD)

↳ Present b/w polar Molecules (HCl & HCl)

(B) Dipole-induced dipole (Debye force)

↳ present b/w polar & Non-polar Molecules. [H_2O, Xe & Br_2, H_2O]

(C) LDf or IDID [London dispersion force or Instantaneous dipole-induced dipole]

↳ present b/w non-polar molecules. [H_2,H_2 F_2,F_2 Xe,Xe Cl_2,Cl_2]

Strength 1 > A > 2 > B > C

ID DD IID DID LDf or IDID

Interaction force (3,4,5,7,7): $f \propto \frac{1}{r^3}$ $\propto \frac{1}{r^4}$ $\propto \frac{1}{r^5}$ $\propto \frac{1}{r^7}$ $\propto \frac{1}{r^7}$

Interaction energy (2,3,4,6,6): $E \propto \frac{1}{r^2}$ $\propto \frac{1}{r^3}$ $\propto \frac{1}{r^4}$ $\propto \frac{1}{r^6}$ $\propto \frac{1}{r^6}$

T to B ↓ M.W↑, Vwf↑, London force↑